

Cauchy-Goursat theorem

If $f(z)$ analytic at all points inside and on a simply closed curve c



then $\int_c f(z) dz = 0$

Cauchy integral formula

If $g(z)$ analytic within and on a closed curve c and if a is any point within c

Evaluate

$$\int_c \frac{z^3}{z-2} dz$$

c

$$|z|=1$$

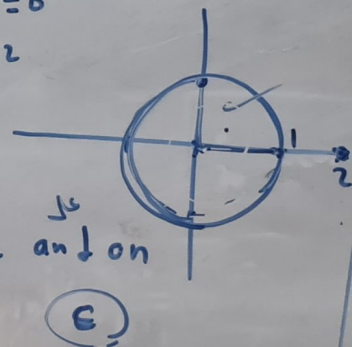
$$z-2=0 \\ z=2$$

$$a=2$$

$$\frac{z^3}{z-2}$$

analytic

inside and on c



$$\int_c \frac{z^3}{z-2} dz = 0$$

$$|z|=1$$

$$|x+iy|=1$$

$$\sqrt{x^2+y^2}=1$$

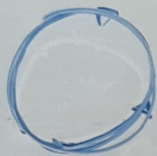
$$x^2+y^2=1$$

circle

$$r=1 \quad (0,0)$$

Cauchy-Goursat theorem

If $f(z)$ is analytic at all points inside and on a simply closed curve C then $\int_C f(z) dz = 0$



Cauchy integral Formula

If $g(z)$ is analytic within and on a closed curve C and if a is any point within C

Evaluate

$$\int_C \tan(z) dz$$

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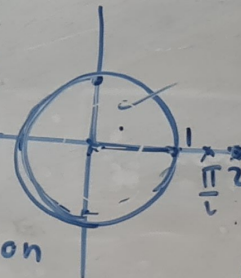
$$|z|=1$$

$$\tan(z) = \frac{\sin(z)}{\cos(z)}$$

$\tan(z)$ is analytic inside and on C

$$\int_C \tan(z) dz = 0$$

$$\begin{aligned} \cos(z) &= 0 \\ z &= \frac{\pi}{2}, \frac{3\pi}{2} \\ z &= -\frac{\pi}{2} \end{aligned}$$



C

$$\frac{\pi}{2} = \frac{3.14}{2}$$

$$|z|=1$$

$$|x+iy|=1$$

$$\sqrt{x^2+y^2}=1$$

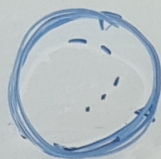
$$x^2+y^2=1$$

center

$$r=1 \quad (0,0)$$

Cauchy-Goursat theorem

If $f(z)$ analytic at all points inside and on a simply closed curve c then $\int_c f(z) dz = 0$



Cauchy integral formula

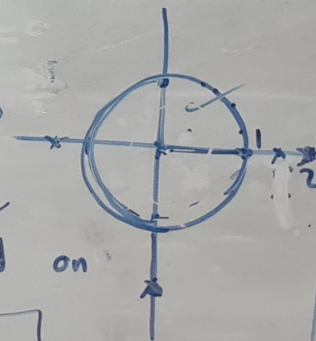
If $g(z)$ analytic within and on a closed curve c and if a is any point within c

Evaluate

$$\int_c \frac{z}{(z-2)(z+2i)} dz$$

$$|z|=1$$

$$\begin{aligned} z-2 &= 0 \\ z &= 2 \\ z+2i &= 0 \\ z &= -2i \end{aligned}$$



$\frac{z}{(z-2)(z+2i)}$ analytic inside and on c

$$\int_c \frac{z}{(z-2)(z+2i)} dz = 0$$

$$\frac{3.14}{2} = 1.57$$

$$|z|=1$$

$$|x+iy|=1$$

$$\sqrt{x^2+y^2}=1$$

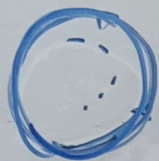
$$x^2+y^2=1$$

$$r=1$$

$$(0,0)$$

Cauchy-Goursat theorem

If $f(z)$ analytic at all points inside and on a simply closed curve C



then $\int_C f(z) dz = 0$

Cauchy integral formula

If $g(z)$ analytic within and on a closed curve C and

if a is any point within C

$$\int_C \frac{g(z)}{z-a} dz = 2\pi i \left[g(z) \right] \Big|_{z=a}$$

$$= 2\pi i g(a) \quad z=a$$

Evaluate

$$\int_C \frac{z^3}{(z-1)(z-3)} dz$$

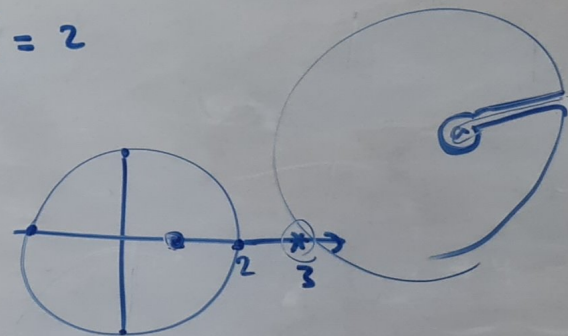
$$C \quad |z|=2$$

$$z-1=0$$

$$z=1$$

$$z-3=0$$

$$z=3$$



$$\int_C \frac{\frac{z^3}{z-3}}{z-1} dz$$

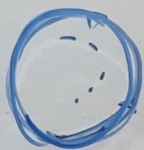
$$\frac{z^3}{z-3}$$

analytic inside and on C

$$\int_C \frac{\frac{z^3}{z-3}}{z-1} dz = 2\pi i \left[\frac{z^3}{z-3} \right] \Big|_{z=1} = 2\pi i \left(\frac{1}{1-3} \right) = 2\pi i \left(\frac{1}{-2} \right) = -\pi i$$

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Cauchy integral formula

If $g(z)$ is analytic within and on a closed curve C and

if a is any point within C

$$\int_C \frac{g(z)}{z-a} dz = 2\pi i \left[g(z) \right]_{z=a} = 2\pi i g(a)$$

Evaluate

$$\int_C \frac{z^3}{z^2-1} dz$$

$$C: |z|=2$$

0

$$|1|=2$$

$$1 \neq 2$$



$$|1|=2$$

$$1 = 2$$

$$\int_C \frac{z^3}{(z-1)(z+1)} dz$$

$$z-1=0 \quad | \quad z+1=0$$

$$z=1$$

$$z=-1$$

$$= \int_{C_1} \frac{z^3}{z-1} dz + \int_{C_2} \frac{z^3}{z+1} dz$$

$$= 2\pi i \left[\frac{1}{z} \right] + 2\pi i \left[\frac{-1}{z} \right] = \pi i + \pi i = 2\pi i$$

